# Solving Linear Equations

## Keep the equation balanced

1. **Distribute** to clear parentheses. Watch the negative!
2. **Combine like terms** on each side.
3. **Variables to one side**: add or subtract a variable term.
4. **Undo + and −** to get the variable term alone.
5. **Undo × and ÷**. Whatever you do to one side, do to the other.
6. **Check**: substitute your answer into the original equation.

### Example

| | |
|---|---|
| $3(x-2)+4=2x+7$ | Start |
| $3x-6+4=2x+7$ | Distribute the 3 |
| $3x-2=2x+7$ | Combine like terms |
| $x-2=7$ | Subtract 2x from both sides |
| $x=9$ | Add 2 to both sides |

Check: $3(9-2)+4=25$ and $2(9)+7=25$ ✓ **same value**

# One, None, or Infinite?

## How many solutions?

Solve until the variable is gone or alone. What is left tells you the answer.

### One solution

$4x + 1 = 2x + 9$

$2x = 8$

$x = 4$

Different *x*-coefficients.

### No solution

$2x + 3 = 2x + 5$

$3 = 5$

**False ✗**

Same *x*-coefficients, different constants.

### Infinite solutions

$2(x + 3) = 2x + 6$

$6 = 6$

**Always true ✓**

Both sides are the same expression.

| What you get | Number of solutions | Graph of each side |
|---|---|---|
| $x =$ a number | One | Lines cross once |
| A false statement, like $3 = 5$ | None | Parallel lines |
| A true statement, like $6 = 6$ | Infinitely many | Same line |

> Same *x*-coefficient on both sides? Compare the constants.

# Literal Equations

## Solve for one letter

A **literal equation** has two or more variables, like a formula. To solve for one letter, treat every other letter as if it were a number and use the same steps.

### Perimeter: solve for $w$

| | |
|---|---|
| $2l + 2w = P$ | Start |
| $2w = P - 2l$ | Subtract $2l$ |
| $w = \frac{P - 2l}{2}$ | Divide by 2 |

### Distance: solve for $t$

| | |
|---|---|
| $d = rt$ | Start |
| $\frac{d}{r} = t$ | Divide by r |
| $t = \frac{d}{r}$ | Rewrite |

### Check with numbers

$l = 5$, $w = 3$ gives $P = 16$. Then $w = (16 - 10) \div 2 = 3$ ✓

# Rewrite Equations

## Divide EVERY term

### Standard form → slope-intercept form

| | |
|---|---|
| $3x + 2y = 8$ | Start |
| $2y = -3x + 8$ | Subtract 3x |
| $y = \frac{-3x}{2} + \frac{8}{2}$ | Divide **every** term by 2 |
| $y = -\frac{3}{2}x + 4$ | Simplify |

### Temperature: solve for $F$

| | |
|---|---|
| $C = \frac{5}{9}(F - 32)$ | Start |
| $\frac{9}{5}C = F - 32$ | Multiply by the reciprocal $\frac{9}{5}$ |
| $F = \frac{9}{5}C + 32$ | Add 32 |

### Undo in reverse order

Undo + and − first, then × and ÷. It is the order of operations, backward.

### Watch out

✗ $2y = -3x + 8 \rightarrow y = -3x + 4$

Only one term was divided. Divide **every** term.

# Inequality Symbols

## Words → symbol → graph

| Symbol | Say it | Key words | Circle |
|---|---|---|---|
| $<$ | less than | fewer than, below | open ○ |
| $>$ | greater than | more than, above | open ○ |
| $\leq$ | less than or equal to | **at most**, no more than, maximum | closed ● |
| $\geq$ | greater than or equal to | **at least**, no less than, minimum | closed ● |

$x < 2$




Open circle: 2 is **not** a solution.

$x \geq -3$




Closed circle: $-3$ **is** a solution.

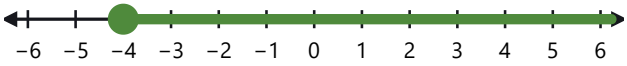
> Variable on the left? The symbol points the way the arrow goes: $x < 2$ shades left, $x \geq -3$ shades right.

# Solving Inequalities

**× or ÷ by a negative → FLIP**

## Example

| | |
|---|---|
| $-3x + 5 \leq 17$ | Start |
| $-3x \leq 12$ | Subtract 5 |
| $x \geq -4$ | Divide by $-3$: **flip** $\leq$ to $\geq$ |




## Why flip?

$2 < 5$ is true.

Multiply both by $-1$:

$-2 \; ? \; -5$

$-2 > -5$, so the sign **flips**.

## Modeling: "at least" means ≥

You earn \$50 plus \$8 for each shirt you sell. You need **at least** \$250.

$50 + 8s \geq 250 \rightarrow 8s \geq 200 \rightarrow s \geq 25$ shirts

Check a point from the shaded side: $x = 0$ gives $5 \leq 17$ ✓ **true**